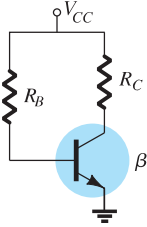
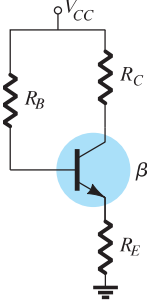
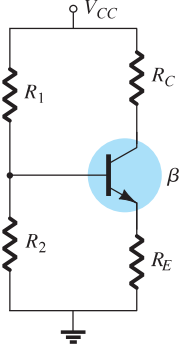
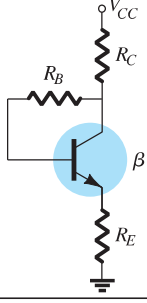
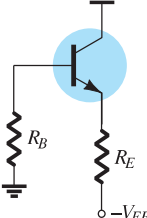
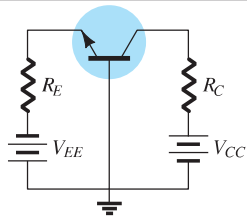


TABLA 4.1

Configuración de polarización del BJT

Tipo	Configuración	Ecuaciones correspondientes
Polarización fija		$I_B = \frac{V_{CC} - V_{BE}}{R_B}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C R_C$
Polarización de emisor		$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $R_i = (\beta + 1)R_E$ $V_{CE} = V_{CC} - I_C (R_C + R_E)$
Polarización por medio del divisor de voltaje		EXACTA: $R_{Th} = R_1 \parallel R_2, E_{Th} = \frac{R_2 V_{CC}}{R_1 + R_2}$ $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C (R_C + R_E)$ APROXIMADA: $\beta R_E \geq 10R_2$ $V_B = \frac{R_2 V_{CC}}{R_1 + R_2}, V_E = V_B - V_{BE}$ $I_E = \frac{V_E}{R_E}, I_B = \frac{I_E}{\beta + 1}$ $V_{CE} = V_{CC} - I_C (R_C + R_E)$
Realimentación de colector		$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{CC} - I_C (R_C + R_E)$
Emisor seguidor		$I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E}$ $I_C = \beta I_B, I_E = (\beta + 1)I_B$ $V_{CE} = V_{EE} - I_E R_E$
Base común		$I_E = \frac{V_{EE} - V_{BE}}{R_E}$ $I_B = \frac{I_E}{\beta + 1}, I_C = \beta I_B$ $V_{CE} = V_{EE} + V_{CC} - I_E (R_C + R_E)$ $V_{CB} = V_{CC} - I_C R_C$